

# DECODED EXPLAINED

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22nd June, 2026

# Yoga Has Become World's Largest Celebration: Modi

## Yoga has become world's largest celebration: Modi

The ancient practice plays a crucial role in promoting physical health, mental well-being, and active ageing, says Prime Minister while participating in International Day of Yoga event in Kolkata

**Shiv Sahay Singh**  
KOLKATA

**P** rime Minister Narendra Modi on Sunday said yoga united and brought everyone together, and had become the world's largest community celebration.

"June 21, in some parts of earth marks the longest day, and because of International Yoga Day, this day has also become the day of the world's largest collective celebration," the Prime Minister said while participating in an event here to mark the occasion.

Mr. Modi, who participated in yoga along with 35,000 people on Red Road here, referred to Ramakrishna Paramhansa, Swami Vivekananda, Mahatma Gandhi, and Lahiri Mahasaya, stating that they took the "yoga tradition to new heights".

Observing that the sense of connection is the very essence of yoga, the Prime Minister said the whole country and the entire world seemed connected to one another, which was the true power of yoga.

In September 2014, Mr. Modi proposed at the United Nations that June 21 be celebrated as International Yoga Day.

On December 11, 2014, the UN officially declared



**Leading the way:** Prime Minister Narendra Modi joined 35,000 people at an event in Kolkata. DPR/PMO

June 21 the International Day of Yoga, with support from 193 member states and 173 co-sponsors. Since then, Yoga Day is celebrated worldwide every year.

### 'Healthy ageing'

In Kolkata, the Prime Minister was seen interacting with the people and correcting the postures of some of them.

Referring to this year's Yoga Day theme, "Yoga for healthy ageing", Mr. Modi said the ancient practice played a crucial role in promoting physical health, mental well-being, and active ageing, thereby improving the quality of life.

Highlighting the bene-

fits of yoga, the Prime Minister said the practice taught people to be lifelong learners of body and mind.

"Our goal should be to be more flexible at 40 than we were at 20; more energetic at 50 than we were at 30. It helps tune our bodies for flexibility, maintains high energy level," he said.

The Prime Minister said people should not restrict yoga to only particular occasions, and they must make it part of their lives.

"Let us pledge not to limit yoga to just one day or a single event. Let us make yoga an integral part of our lives, our families, and the lives of future genera-

tions," the Prime Minister said.

### Grand event

This was the first time Yoga Day celebrations were held in Kolkata with a lot of fanfare. West Bengal Chief Minister Suvenudh Adhikari and several other BJP leaders and Ministers were present. The venue and the Kolkata riverfront were decked up over the past several days. Screens live-streaming the event were set up in several parts of the city.

"Celebrating International Yoga Day in such a vibrant city is truly an honour," the Prime Minister posted on social media.

# **Yoga Has Become World's Largest Celebration: Modi**

## **● International Day of Yoga (IDY)**

- **Observed every year on June 21 since 2015**
- **Proposed by PM Modi during his address at the 69th UN General Assembly, September 2014**
- **UN officially declared June 21 as IDY on December 11, 2014**
- **Supported by 193 UN member states and 173 co-sponsors**

## **● 2026 Theme: “Yoga for Healthy Ageing”**

- **Main event held for the first time in Kolkata; PM Modi joined 35,000 people on Red Road**
- **PM said yoga teaches people to be “lifelong learners of body and mind”**

## **● Significance of June 21**

- **Marks the Summer Solstice : the longest day in parts of the Northern Hemisphere**

## **● India's Yoga Tradition**

- **PM referred to Ramakrishna Paramhansa, Swami Vivekananda, Maharshi Aurobindo and Lahiri Mahasaya as having taken the “yoga tradition to new heights”**

# A 10 Percent Rule for Sustainable Development



RAJESH GOPAL

INDIA'S DEVELOPMENT trajectory is entering a decisive phase. Infrastructure expansion, renewable energy transitions and economic growth are accelerating, often in the landscapes that underpin the country's ecological security. Forested regions, many overlapping with tiger habitats and wildlife corridors, are increasingly at the centre of this transformation. This raises a difficult question: How much ecological cost is too much?

For decades, India has relied on regulatory instruments such as Environmental Impact Assessments (EIA), forest clearances and compensatory afforestation to manage this balance. More recently, the idea of Green

GDP — adjusting economic output to reflect environmental loss — has gained attention. Yet these approaches are often difficult to translate into clear decision thresholds. Policymakers need not merely more data but a simple, credible benchmark to recognise when economic activity begins to undermine its own ecological foundation. An answer may lie in a principle that governs all natural systems. In every ecosystem, energy flows through a hierarchy — from plants to herbivores to predators. At each step, most of the energy is lost, leaving only a small fraction available for the next level. This principle, described by Lindeman's 10 per cent Law, explains why nature forms pyramids.

In much of India's forests, the tiger is at the pyramid's apex. Its survival depends on a large and stable prey base, which in turn depends on healthy vegetation, water sys-

tems and minimal disturbance. Even small disruptions at the base can cascade upward, destabilising the system. This ecological reality offers an important economic insight: Systems can absorb only limited stress before they begin to degrade.

Recent estimates place the economic cost of environmental degradation in India at around 9-11 per cent of GDP. A precautionary approach would be to treat this as an eco-

Recent estimates place the economic cost of environmental degradation in India at around 9-11 per cent of GDP. A precautionary approach would be to treat this as an ecological threshold

logical threshold — a point beyond which the economy is not operating sustainably and is drawing down its natural capital. Development must pay a minimum ecological due.

This idea acquires particular urgency in tiger landscapes. India's tiger reserves and corridors are embedded within landscapes that support agriculture, infrastructure and local livelihoods. They also provide critical ecosystem services, including water regulation, carbon storage and biodiversity conservation. These systems also operate under tight ecological margins. Incremental pressures — linear infrastructure, mining, unsustainable extraction and land-use change — can accumulate rapidly. The consequences are already visible in many regions: Rising human-wildlife conflict, fragmentation of corridors and stress on prey populations.

In such contexts, a simple threshold can

serve as an early warning. If the cumulative ecological cost of activities within a landscape begins to exceed a notional 10 per cent of its total economic or ecological value, it should trigger a policy pause — not to halt development, but to reassess its scale, location and design. In more fragile wildlife corridors, an even more conservative threshold of 5-8 per cent may be appropriate. Such an approach can be incorporated into EIA processes, landscape-level planning in notified corridors, green accounting frameworks and infrastructure design.

In nature, as in economics, systems that consume their foundations do not collapse immediately. But they do, eventually — and often without warning.

*The writer is secretary general, Global Tiger Forum*

# A 10 Percent Rule for Sustainable Development

## ● The Problem

- India's infrastructure, renewable energy and economic growth often overlap with ecologically sensitive landscapes, especially tiger habitats and corridors
- Existing tools : EIAs, forest clearances, compensatory afforestation, Green GDP → lack a simple, actionable threshold

## ● Lindeman's 10 Percent Law

- Ecological principle: only about 10% of energy is transferred from one trophic level to the next; most is lost as heat
- Explains why ecological pyramids narrow toward the top the tiger sits at the apex, depending on a large, stable prey base

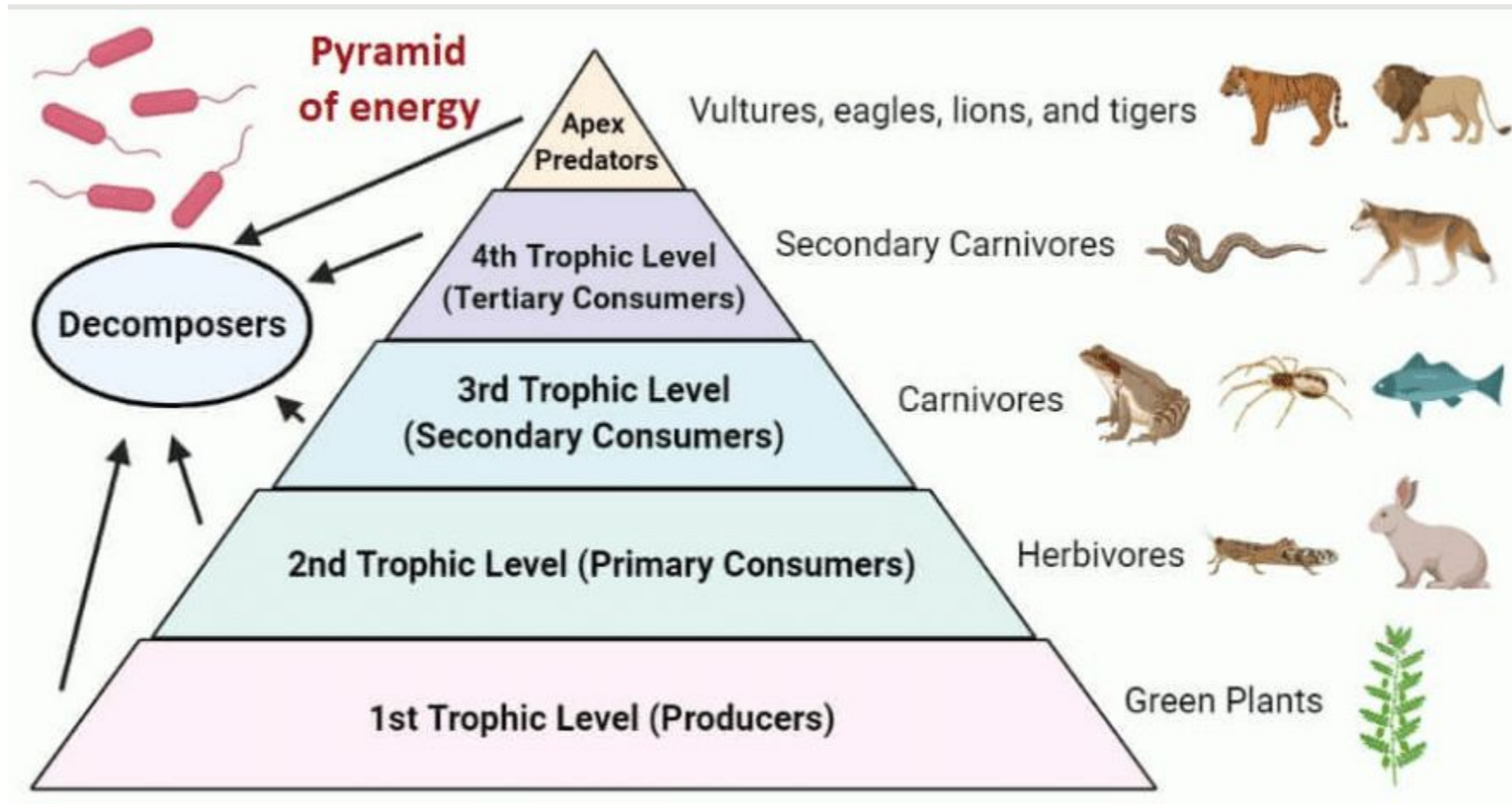
## ● Proposed Economic Threshold

- Recent estimates place the economic cost of environmental degradation in India at around 9–11% of GDP
- A precautionary “10% rule”: if cumulative ecological cost of development exceeds ~10% of a landscape's total economic/ecological value, it should trigger a policy pause to reassess scale, location and design, not to halt development
- In more fragile wildlife corridors, a stricter 5–8% threshold may be appropriate

## ● Application

- Can be built into EIA processes, landscape-level planning in notified corridors, green accounting frameworks, and infrastructure design

## A 10 Percent Rule for Sustainable Development



## **A 10 Percent Rule for Sustainable Development**

### **Practice Question (PYQ-style)**

**Consider the following statements regarding Lindeman's 10 Per Cent Law:**

- 1. It states that approximately 10% of energy at one trophic level is transferred to the next trophic level.**
- 2. It explains why ecological pyramids of energy are typically broader at the top than at the base.**
- 3. It has been suggested as an analogy for setting a precautionary threshold for the permissible ecological cost of development.**

**Which of the statements given above is/are correct?**

- (a) 1 only**
- (b) 1 and 3 only**
- (c) 2 and 3 only**
- (d) 1, 2 and 3**

## A 10 Percent Rule for Sustainable Development

### Answer

**Correct Option: b) 1 and 3 only**

### **Explanation:**

**Statement 2 is incorrect — ecological pyramids narrow (not broaden) toward the top, because energy is progressively lost at each trophic transfer.**

# Changed Reality: India Must Reduce Its Dependence on the Strait of Hormuz

## Changed reality

India must reduce its dependence on the Strait of Hormuz

Nations rise to power on the strength of their economies and, by extension, trade. Historically, countries such as the United Kingdom, Japan, the United States and China have combined economic prosperity with maritime dominance. India's weak shipping sector reflects its geopolitical position. Yet, its seafarers generate billions of dollars in foreign exchange while facing risks from piracy and geopolitical tensions. The recent conflict with Iran showed that control of strategic waterways can be as consequential as economic sanctions. By leveraging the Strait of Hormuz, Iran exposed a critical vulnerability in the global economy. Disruptions to energy flows imposed costs far beyond the battlefield and the economic consequences appear to have played a significant role in shaping the U.S. response. From a shipping perspective and its Hormuz stakes, Iran may have secured strategic gains despite the heavy damage inflicted on it by Israeli and U.S. Even as the attacks on Lebanon cast a shadow over the peace deal, Iran has announced that the Persian Gulf Strait Authority it set up during the war will be the sole authority that will handle Strait of Hormuz transits.

The memorandum of understanding envisages lifting sanctions on Iran and on ships serving Iranian trade. It also mandates Iranian talks with Oman and other Gulf states on maritime administration in the Strait of Hormuz. Previously, vessels transiting the Strait neither paid tolls nor reported to Iran or Oman. Whether or not the new framework results in transit fees is less important than the reality: that shipping companies must now account for Iran as a decisive stakeholder in the Strait of Hormuz. For India, the conflict has exposed a major strategic weakness. Despite the Strait of Hormuz being vital to its energy security, the disruption exposed the absence of a credible contingency plan for such eventualities. India's LPG strategy relies heavily on imports moving through the Strait of Hormuz, supported by a limited fleet of Indian-flagged carriers and a tightly scheduled supply chain with little long-term cavern storage. Similar vulnerabilities affected other fuel supplies too. Many countries are now reassessing their dependence on the Strait of Hormuz. The United Arab Emirates is pursuing a "zero Hormuz dependency" strategy by strengthening alternative infrastructure and routes. India must draw lessons from the crisis. Diversifying supply chains, investing in alternative maritime and land corridors, and strengthening strategic partnerships should become priorities. Projects such as Chabahar offered precisely such an opportunity but India gave up on it. For India, reducing dependence on the Strait of Hormuz is no longer just an economic goal; it is a strategic necessity.

# Changed Reality: India Must Reduce Its Dependence on the Strait of Hormuz

## ● Why the Strait Matters

- A critical maritime choke point linking the Persian Gulf to the Gulf of Oman and the wider Indian Ocean
- India's LPG imports and broader energy security depend heavily on transit through the Strait

## ● Recent Conflict & Its Lessons

- The recent Iran conflict showed that control of strategic waterways can be as consequential as economic sanctions
- Iran's leverage over the Strait exposed a critical vulnerability in the global energy economy
- A new “Persian Gulf Strait Authority” set up by Iran during the conflict has been announced as the sole authority for handling Strait of Hormuz transits

## ● New Arrangements

- A memorandum of understanding envisages lifting sanctions on Iran and on ships serving Iranian trade
- It mandates Iranian talks with Oman and other Gulf states on maritime administration of the Strait
- Previously, vessels transiting the Strait neither paid tolls nor reported to Iran or Oman this is now set to change

## ● India's Vulnerability & Way Forward

- India gave up an early opportunity via the Chabahar Port project to diversify access routes
- Way forward: diversify supply chains, invest in alternative maritime/land corridors, strengthen strategic partnerships  
(Example : UAE's “zero Hormuz dependency” strategy)

## Changed Reality: India Must Reduce Its Dependence on the Strait of Hormuz



**Bordering Countries:** Iraq, Kuwait, Saudi Arabia, Bahrain, Qatar, United Arab Emirates, Oman (Musandam exclave) and Iran.

The Strait of Hormuz is situated between **Qeshm Island** and the Iranian coast in the north and the **Musandam Peninsula** of the Arabian Peninsula in the south.

To the west, it connects to a large river delta called **Shatt al-Arab**. Here, the waters from two major rivers, the **Euphrates and the Tigris**, flow into the Gulf.

## Changed Reality: India Must Reduce Its Dependence on the Strait of Hormuz

### BEING DIRECT: INDIA TO CHABAHAR



### Chabahar Project

In May 2016, India, Iran, and Afghanistan signed a trilateral agreement to establish the International Transport and Transit Corridor, also known as the Chabahar Agreement.

It is located on the Gulf of Oman and is only 72 km away from the Gwadar port in Pakistan which has been developed by China.

It is situated in Sistan and Baluchistan Province, on the Makran coast. It lies in the Gulf of Oman.

It is the only Iranian port with direct access to the Indian ocean and consists of two separate ports named Shahid Beheshti and Shahid Kalantari.

## **Changed Reality: India Must Reduce Its Dependence on the Strait of Hormuz**

### **Practice Question (PYQ-style)**

**The Strait of Hormuz, sometimes seen in the news, connects which of the following two water bodies?**

- (a) Red Sea and the Mediterranean Sea**
- (b) Persian Gulf and the Gulf of Oman**
- (c) Gulf of Aden and the Arabian Sea**
- (d) Persian Gulf and the Suez Canal**

## **Changed Reality: India Must Reduce Its Dependence on the Strait of Hormuz**

### **Answer**

**Correct Option: b) Persian Gulf and the Gulf of Oman**

### **Explanation:**

**The Strait of Hormuz is the narrow waterway connecting the Persian Gulf to the Gulf of Oman and onward to the Arabian Sea, making it one of the world's most strategically important energy chokepoints.**

# Three Distinct Strategic Roles of the Three Ships Inducted into Navy

## Three distinct strategic roles of the three ships inducted into Navy

Sushant Kulkarni

Pune, June 21

THE NAVY inducted three ships on Sunday — one that can fight far out at sea, one that can map the sea, and one that can hunt submarines close to the coast.

The three vessels — the Brahmos-armed stealth frigate INS Dunagiri, the deep-water survey vessel INS Sanshodhak, and the Anti-Submarine Warfare Shallow Water Craft (ASW SWC) INS Agray — have been built by Garden Reach Shipbuilders & Engineers (GRSE) in Kolkata. According to the Ministry of Defence, the three together have more than 75 per cent indigenous content, and involve over 200 MSMEs.

### What are these three ships?

**INS DUNAGIRI:** The largest and most heavily armed of the three is INS Dunagiri. It is a frigate, smaller than a destroyer but large enough to operate far from the coast.

It is part of the Navy's Project 17A, under which a new generation of stealth guided-missile frigates are being built in India. Dunagiri carries weapons such as BrahMos surface-to-surface missiles, sensors including Multi-Function Surveillance, Track



PM Modi presided over the commissioning ceremony in Kolkata on Sunday. PTI

And Guidance Radar (MFSTAR), electronic warfare systems and anti-submarine weapons. "At the strategic level, these multi-mission frigates are capable of operating in a 'blue water' environment — deep sea far from shore — dealing with both conventional and non-conventional threats," a serving Navy officer said. Other vessels in the class are INS Nigiri, Himgiri, Taragiri, Udaygiri and Vindhyagiri.

**INS SANSHODHAK:** This is a Survey Vessel

— Large (SVL). Its job is to measure and map the sea: the depth of waters, seabed features, approach channels to ports, navigational routes, and oceanographic data. It is equipped with systems such as autonomous underwater vehicles, remotely operated vehicles and multi-beam echo sounders. These help the ship collect data from the surface and underwater.

"Warships and submarines do not operate in an empty blue space. They move through waters shaped by depth, seabed features, currents, reefs, channels, ports, underwater slopes and coastal clutter. Knowing this environment helps ships move safely, submarines plan routes, ports update charts, and forces prepare for operations. It also supports civilian needs such as safe shipping, disaster relief, ocean research and coastal development," said a Navy officer who has served on survey ships.

**INS AGRAY:** The smallest vessel of the three, it has a highly specialised role. Agray belongs to the Arnala-class of smaller warships built specifically to detect and attack submarines in shallow waters near the coast, ports, naval bases and important sea approaches. Agray carries lightweight torpedoes, indigenous anti-submarine rocket

launchers and sonar systems.

"Coastal waters are difficult places to find submarines. They are noisy, busy and cluttered. Fishing boats, merchant ships, seabed features and coastal activity can make submarine detection harder than in the open ocean. That is where ASW-SWCs such as Agray come in," said an officer.

### What is the larger message behind commissioning them together?

The importance of Sunday's ceremony is that the three ships add three different capabilities at once.

As the Indian Ocean becomes more contested, the triple-commissioning "is a sign of how the Navy is building capability in layers: big warships for distant operations, survey ships to understand the sea, and smaller submarine-hunters to guard coastal waters," said an officer.

It is also an industrial signal. That three different kinds of ships, with different missions and technologies, can be built domestically and commissioned together shows the growing maturity of India's naval shipbuilding ecosystem.

FULL REPORT ON

[WWW.INDIANEXPRESS.COM/EXPLAINED](http://WWW.INDIANEXPRESS.COM/EXPLAINED)

## Three Distinct Strategic Roles of the Three Ships Inducted into Navy

Frigates are usually regarded as ships weighing more than 3000 tons. Their role is to protect other ships in their strike group, the main part of this responsibility is to protect them from hostile submarines.

A Corvette is a small warship with light arms. Due to this and their high maneuverability they are great in smaller seas

Destroyers are among the widely used ships in the world due to their wide range of use.



## Three Distinct Strategic Roles of the Three Ships Inducted into Navy

**Project 17A:** P17A ships feature enhanced stealth capabilities compared to their predecessors P17 (Shivalik) class, with improvements in hull design and weapon systems.

The seven ships under Project 17A, are **INS Nilgiri, INS Udaygiri, INS Himgiri, INS Taragiri, INS Mahendragiri, INS Dunagiri, and INS Vindhyagiri.**

### **● Three Ships, Three Roles**

- **INS Dunagiri :** BrahMos-armed stealth frigate, part of **Project 17A**; operates in 'blue water' (deep sea, far from coast)
- **INS Sanshodhak :** Survey Vessel (Large); maps depth, seabed features, navigational routes and oceanographic data
- **INS Agray :** Anti-Submarine Warfare Shallow Water Craft (ASW SWC) of the Arnala-class; detects and attacks submarines in shallow coastal waters

### **● Built by GRSE, Kolkata**

- **Over 75% indigenous content; involved more than 200 MSMEs** reflects Atma Nirbhar Bharat in defence manufacturing

## **Three Distinct Strategic Roles of the Three Ships Inducted into Navy**

### **Practice Question (PYQ-style)**

**With reference to recent inductions into the Indian Navy, consider the following pairs:**

- 1. INS Dunagiri - Stealth guided-missile frigate**
- 2. INS Sanshodhak - Anti-Submarine Warfare Shallow Water Craft**
- 3. INS Agray - Deep-water Survey Vessel**

**How many of the pairs given above are correctly matched?**

- (a) Only one pair**
- (b) Only two pairs**
- (c) All three pairs**
- (d) None of the pairs**

## Three Distinct Strategic Roles of the Three Ships Inducted into Navy

### Answer

**Correct Option: a) Only one pair**

### **Explanation:**

**Only pair 1 is correctly matched. INS Sanshodhak is actually a Survey Vessel (Large), and INS Agray is the Anti-Submarine Warfare Shallow Water Craft — the roles in pairs 2 and 3 are swapped.**

# End the Free Rein of Junk Food Advertising in India

## End the free rein of junk food advertising in India

Despite the Government of India's plans to amend advertising laws to curb the promotion of HFSS (high in fat, sugar and sodium) foods, such products continue to be advertised rampantly. As evidence of the health harms associated with industrially processed foods engineered to be highly palatable and potentially addictive continues to grow, restricting their advertising – particularly exposure to children and young people – may no longer be avoidable.

Try opening a YouTube video on politics, scrolling through Instagram reels, or scanning a newspaper, and you are likely to encounter advertisements for noodles, chips, biscuits, breakfast cereals, chocolates, sweetened beverages, or other ultra-processed food (UPF) products. Recently, there was a YouTube advertisement for a newly launched baked chips brand in India. The advertisement emphasised the product's cheese and tomato flavours and the "crunchiness" to appeal to consumers. What it did not disclose was that the product is a UPF with ingredients such as maltodextrin, nature-identical flavourings, flavouring substances, salt substitute (KCl/potassium chloride), acidity regulators (627, 631) and emulsifier (322). While prominently promoting selective attributes such as "baked", the advertisement omitted material health information, including the product's high salt and fat content and the presence of refined carbohydrates. Such marketing practices can create a misleading impression of healthfulness while obscuring the nutritional risks associated with these products.

While readers can recall their own experiences, there are a few other examples in the media. A female film celebrity is seen recommending a multigrain, "no maida choco cereal" for her son, despite it being a high sugar product. An entire family of actors promotes a "12-grain" breakfast cereal, while a popular film actor endorses a biscuit as a "good choice". Most of these products, however, are high in sugar, fat and/or salt, raising questions about the messages conveyed through such endorsements. Such selective disclosures create a false perception of healthfulness and deprives consumers, particularly children and adolescents, of the right to make an informed choice.

### Review frameworks

The focus of this article is also to draw the attention of policymakers to the need for reviewing whether existing legal frameworks sufficiently serve the public interest. Clearer legal provisions may be required to effectively regulate the advertising of unhealthy food products.

Advertising is directly linked to increased consumption of UPFs, which is strongly associated with rising rates of obesity and diabetes. These advertisements often feature



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Restricting the advertising of ultra-processed food (UPF) products and foods high in fat, sugar and sodium (HFSS) is a public health imperative

child actors and use emotionally appealing messages aimed at both children and parents, creating a desire for such products. The fact that in 2024, three major transnational corporations spent \$13.2 billion on advertising underscores the volumes and the power of food product advertisements. Advertising does not merely reflect demand; it helps create it. In India alone, more than two lakh junk food advertisements in a month were backed by advertising expenditure of about ₹170 crore.

Evidence suggests that UPFs can encourage overconsumption through mechanisms that resemble those identified in addiction science. The health harms associated with UPFs appear closely linked to their industrial design and marketing strategies. But the food industry fails to disclose this fact to people. Recently, the City of San Francisco filed a lawsuit against 10 major UPF manufacturers, alleging child-targeted marketing, the development of highly compelling product formulations, and inadequate disclosure of health risks such as obesity and diabetes. Among other remedies, the lawsuit sought to prevent further deceptive marketing practices and pushed for corrective measures to address the effects of past false advertising.

### A policy gap

The Government of India's National Multisectoral Action Plan (NMAP) for Prevention and Control of Common Non-Communicable Diseases (2017-2022) envisaged the prohibition/restrictions on the advertising of HFSS foods. Many pre-packaged foods are highly processed, containing additives such as colours, flavours, emulsifiers and sweeteners, and are often HFSS. The issue has gained policy attention. In February 2026, the Supreme Court of India, in response to a PIL on warning labels for packaged foods, observed that front-of-pack labelling is necessary to protect the right to health. The Economic Survey 2025-26 also highlighted concerns around unhealthy diets. Several Members of Parliament have called for stronger measures, including front-of-pack warning labels, advertising restrictions and taxation of UPFs. In 2024, the Court had noted that misleading advertisements can encourage the consumption of unhealthy foods by children, pregnant women and the elderly, with potentially serious health consequences. These developments point to a growing recognition that existing safeguards may be inadequate.

The Lancet Series on UPFs and Human Health, published three papers in November 2025 which presented scientific evidence linking UPF consumption to poorer diet quality, displacement of real foods, and a higher risk of obesity, hypertension, cardiovascular disease, type 2 diabetes and other non-communicable diseases. Global and Indian data show that rising UPF consumption has coincided with increasing

obesity rates. The Lancet made a strong case for policies in the food environments to reduce UPF consumption, with many experts arguing that policymaking should not wait for further evidence.

### The food environment needs a fix

Children and adolescents in India are exposed daily to advertisements for UPFs and HFSS foods on television, digital platforms, social media, sports broadcasts and through influencers. This sustained and sophisticated marketing is designed to build brand loyalty and shape lifelong consumption patterns. The aim of the UPF industry is clear: to encourage the displacement of real culinary or cultural foods for profits. What children or youth eat cannot be separated from what they are persuaded to desire at schools, work places, cinema halls, other public places or even at home.

Experts in The Lancet Series contend that nutrition education and behaviour-change programmes alone cannot succeed in an environment that is saturated with aggressive marketing of unhealthy food products.

This situation underscores an important constitutional principle: when harm is foreseeable and populations are vulnerable, the state has a duty to protect public health and regulate the marketing of unhealthy food products. India committed in 2017 to restrict such advertising, but that objective remains unfulfilled. Given the scale of the problem, neither market forces nor self-regulation are likely to be sufficient. Therefore, there is a strong case for the Government of India to introduce stricter controls on the advertising and promotion of UPFs and HFSS foods such as planned in 2017 by amending advertisement laws.

If schools are to be protected spaces free from UPFs, HFSS foods and misleading nutrition messages, it is inconsistent to ignore the commercial environment that shapes children's choices outside school. The school environment itself requires clear policy direction, not merely advisories (as Brazil did recently). The Economic Survey has called for stronger regulation of UPF advertising and marketing. International experience, from Chile to Mexico, suggests that voluntary self-regulation is often ineffective, whereas enforceable legal measures can be more effective. Given its influence on children's food choices, advertising warrants stronger regulation as part of the broader right to health.

Restricting the advertising of unhealthy food products need not be viewed as anti-industry or anti-profit. In fact it could reduce company expenditure on advertising and encourage companies to redirect resources towards minimally processed foods and healthier local markets. Such a shift could help shape more sustainable and health-oriented food systems in the future.

# End the Free Rein of Junk Food Advertising in India

## ● The Problem

- HFSS (High in Fat, Sugar, Sodium) and Ultra-Processed Food (UPF) products continue to be advertised widely despite government plans to amend advertising laws
- Advertising does not merely reflect demand it helps create it; emotionally appealing, child-targeted marketing is common
- In India, more than two lakh junk food advertisements appear monthly, backed by about ₹170 crore in ad spend

## ● Health Linkages

- UPF consumption can encourage overconsumption through mechanisms resembling addiction science
- The Lancet Series (Nov 2025) links UPFs to poorer diet quality and higher risk of obesity, hypertension, cardiovascular disease, type 2 diabetes and other NCDs

## ● Policy Developments

- National Multisectoral Action Plan (NMAP) for NCDs (2017–2022) had envisaged restrictions on HFSS advertising
- Feb 2026: Supreme Court, responding to a PIL on warning labels, observed that front-of-pack labelling is necessary to protect the right to health
- Economic Survey 2025–26 flagged concerns around unhealthy diets and called for stronger regulation

## ● The Way Forward

- Stronger legal provisions and enforceable controls rather than voluntary self-regulation on UPF/HFSS advertising
- International examples: Chile and Mexico (enforceable measures); Brazil (school advisories)

# In Time of Distrust, Constitution's Promise of Fraternity Matters More Than Ever

## In time of distrust, Constitution's promise of fraternity matters more than ever



S Y QURAISHI

**E**VERY SCHOOLCHILD can recite the Preamble to the Constitution: Justice, liberty, equality and fraternity. The first three have generated debate in courts, politics and academia. The fourth, however, has passed almost unnoticed. Yet fraternity may be the most consequential of all the constitutional values because it is the one value that law alone cannot create.

The neglect is curious because the Constitution does not treat fraternity as an afterthought. It appears in the same sentence as the other three, carrying equal constitutional weight. The framers placed it there deliberately. The question is why we have spent 75 years largely ignoring it, and what that neglect has cost us.

The answer begins with a distinction that was explicit during the Constituent Assembly debates but has since faded from constitutional discourse. Liberty and equality define the relationship between the citizen and the state. Fraternity defines the relationship between citizens themselves. That distinction makes all the difference. If the state violates liberty or equality, courts can intervene. But what recourse exists when citizens cease to regard one another as fellow citizens? When disagreement hardens into hostility and difference into suspicion, fraternity begins to disappear. B R Ambedkar understood this with unusual clarity. In his final address to the Constituent Assembly on November 25, 1949, he warned, "Without fraternity, equality and liberty will be no deeper than coats of paint." His warning stemmed from a frank assessment of Indian society. "How can people divided into several thousands of castes be a nation?" he asked. He was reminding us that nationhood is not a given but an achievement, and fraternity both its condition and proof. The Constitution's concern with fraternity did not end with the Preamble. Article 14(e) makes it a Fundamental Duty of every citizen to pro-



ILLUSTRATION: C R SASIKUMAR

mote harmony and the spirit of common brotherhood across religious, linguistic and regional differences.

Seventy-five years later, the original anxieties have not diminished. Though we have never been more connected technologically, we have rarely been more estranged socially. Public debate has grown louder, sharper and more vicious, while mutual understanding has become thinner. The greatest damage caused by hate-filled discourse — whether political, religious or digital — is not that it offends. Democracies have always survived offence. Its deeper damage lies in the erosion of trust.

Trust is the invisible infrastructure of a functioning society — the assumption that strangers can deal with one another in good faith. When that assumption erodes, no law can fully restore it. The economy depends upon it. Democracy depends upon it.

Liberty and equality are justiciable, enforceable and actionable before a judge. Fraternity belongs to a different category. If liberty and equality are the fundamental rights of citizens, fraternity, for me, is the fundamental right of the nation. No court can manufacture mutual respect. No statute can create a sense of common belonging. No government programme can legislate trust. Yet it would be wrong to conclude that fraternity lies entirely beyond the reach of law and the government and the courts are helpless spectators.

No court can manufacture mutual respect. No statute can create a sense of common belonging. Yet it would be wrong to conclude that fraternity lies entirely beyond the reach of law

The Constitution and the criminal law contain a substantial architecture designed to protect social harmony. Article 14 guarantees equality before the law. Articles 15 and 25 protect citizens against discrimination and safeguard freedom of faith. Article 14(e) imposes a duty to promote harmony and common brotherhood. Criminal law goes further. Sections 153A, 153B, 295A and 505 of the IPC, along with provisions protecting places of worship, were enacted to preserve civic peace and prevent communal hostility.

These provisions may not create fraternity, but they exist to prevent its systematic destruction. The challenge, therefore, is not the absence of legal safeguards. It is their uneven enforcement. Courts cannot command affection, but they can act against hatred, intimidation and attacks on equal citizenship. The law cannot manufacture fraternity, but it can — and must — prevent its destruction.

It is worth asking what 75 years of neglecting fraternity has produced. Consider what the government's own data tell us. For more than a decade, roughly 2 lakh Indians have been surrendering their citizenship every year, amounting to nearly 9 lakh in the last five years alone, according to information placed before Parliament. Many of those leaving are successful entrepreneurs and professionals. Their decisions reflect multiple factors, but the scale of the phenomenon raises a legitimate question: What role do social climate and civic

confidence play in such choices?

History suggests that prolonged social conflict leaves economic scars that persist long after the headlines disappear. India has already learned this lesson. Punjab was once the most prosperous state in the country, with the highest per-capita income as recently as the mid-1980s. The decade-long militancy and communal discord that followed devastated the investment climate. Punjab slipped from first place in per-capita income rankings and, between 2014-15 and 2022-23, it ranked 18th in economic growth among 21 major states. The damage was not merely economic; poisoned atmospheres take generations to clear.

The irony is difficult to miss. We have spoken of India becoming the world's third-largest economy and a fully developed *Viksit Bharat* by 2047. Yet public institutions have often appeared insufficiently concerned about an atmosphere of anxiety and periodic lawlessness that seems to corrode such ambitions. Lynchings, targeted demolitions and the normalisation of communal hostility are not merely law-and-order failures. They are fraternity failures. When such acts go unpunished, they send a signal — not just to minorities, but to every entrepreneur, investor and family weighing its future.

Polarisation may have delivered electoral dividends to those who have practised it. But a nation cannot continuously inflame its divisions for short-term political advantage and expect to contain its economic consequences.

Fraternity is not a pious wish inscribed in a preamble and ceremonially quoted on national occasions. It is the social oxygen without which neither liberty nor equality can survive, and without which no economy, however ambitious its targets, can truly flourish. Ambedkar warned that without fraternity, liberty and equality would be no deeper than coats of paint. For three-quarters of a century, we barely noticed the paint peeling. We are beginning to see the cracks now.

The framers placed all four values together for a reason. It may be time we did the same.

*Quraishi is former chief election commissioner of India and author of An Undocumented Wonder: The Making of the Great Indian Election*

## **In Time of Distrust, Constitution's Promise of Fraternity Matters More Than Ever**

### **● Fraternity in the Constitution**

- **Appears in the Preamble alongside Justice, Liberty and Equality given equal constitutional weight by the framers**
- **Liberty and equality define the citizen–state relationship; fraternity defines the citizen–citizen relationship**
- **Article 51A(e): Fundamental Duty to promote harmony and the spirit of common brotherhood among all citizens**

### **● Ambedkar's Warning**

- **Final address to the Constituent Assembly (Nov 25, 1949): “Without fraternity, equality and liberty will be no deeper than coats of paint”**

# Scrutiny and Security: How China Conducts Its Biggest College Entrance Exam

## Scrutiny and security: How China conducts its biggest college entrance exam

Rishika Singh  
New Delhi, June 21

THE RE-TEST for the National Eligibility cum Entrance Test (NEET) was held on Sunday, a few weeks after allegations of paper leak led to the original exam's cancellation.

Earlier this month, China also conducted its biggest national exam for college admissions, the Gaokao (literally, "the highest exam"), with over 1.29 crore students. A spokesperson of the Chinese Embassy in India shared a post on X, stating that "The #Gaokao is one of the most important exams for millions of Chinese students — and what makes it remarkable is its fairness... Students know their hard work truly pays off, because the process is strictly regulated and the road to academic success is not rigged."

**Teachers under supervision, no internet access**

Gaokao has been conducted in China

since 1952. During the Mao Zedong-led Cultural Revolution (1966-76), which saw sweeping changes to align Chinese society with Communist ideals, the exam was put on hold.

Today, the exam includes three compulsory subjects: English, Mandarin and Mathematics, with optional science and social science subjects. Chinese state media reports note that the full exam lasts between three and four days, and the duration varies by region.

There are strict protocols for the conduct of the exam, including but not limited to question papers drafted under strict security. Teachers who are selected to prepare the question papers are sent to isolated locations, including military facilities and prisons.

And it's not just the state that goes all out. "Among other heart-warming measures, some companies offer free taxi rides to candidates going to their exam centres, while some restaurants provide free lunches, and some hotels free beds, on gaokao days," a recent Chinese state media report said.

Chinese state media reports also point to attempts to curb unfair practices during exams. A 2025 report said about the province of Liaoning, "Students are required to pass through two intelligent security gates capable of detecting small metal objects, including mobile phones, smartwatches, smart glasses, hidden earpieces and other high-tech devices that could be used to cheat. If an alarm is triggered, teachers follow up with handheld metal detectors."

### Not a foolproof system

It quoted a high school principal as saying, "Cheating technologies are evolving every year, so this year's entry screening process is stricter than ever."

Biometric verification is conducted at entry to detect student impersonation. Cases of impersonation have previously been reported years after the exam, and punishments have been handed out. In 2016, six teachers were handed a jail sen-

tence from one year to 26 months after being convicted of "illegally obtaining state secrets".

In 2025, state media *China Daily* reported that the city of Benxi, with a population of around 1.3 million, had every gaokao room monitored by a "real-time intelligent surveillance system that automatically flags irregular behavior — from whispers and frequent glances between students to inattentive proctors". Other recent reports have also touted "AI-powered" systems.

The exam involves both objective and subjective questions. Experienced teachers from key senior high schools and universities in a region are selected to grade the papers. "Personnel involved in grading have to present their identity documents, leave electronic devices in a small basket in a separate area, and undergo security checks before work starts," an official media report said.

Grading computers are connected to an exclusive network to prevent hacking and intrusion of viruses and the rooms are

equipped with surveillance cameras, with police officers on duty 24/7, it added.

### STEM focus

In their book *The Highest Exam: How the Gaokao Shapes China* (2025), authors Hongbin Li, Ruixue Jia and Claire Cousineau wrote, "If China's education system is a symphony orchestra, the central government is the conductor." It directs everyone, but relies on musicians (in this example, the local governments) to achieve what is desired, they write. Local authorities own and fund non-elite colleges and institutions, as well as schools, but they operate based on policies made by the Central government.

The book credits Deng Xiaoping, the leader under whom the Chinese economy modernised in 1978, for raising the scale and scope of the education system. Among 18 to 24-year-olds, the share of people enrolled in college increased from 1% in 1976 to 57% in 2021.

The exam has come under criticism for

furthering inequalities, particularly on regional and class lines. Colleges can set local quotas and most premier institutions are based in major cities.

Combined with China's household registration system that designates citizens as hailing from rural or urban areas, it means that those in cities have a disproportionately high chance of making it to top colleges. They are also more equipped to spend a considerable amount on additional classes and other resources.

A type of affirmative action system awards some advantages to ethnic minorities, including those from the regions such as Tibet.

Over the years, there has been an increasing focus on pushing Science, Technology, Engineering and Mathematics (STEM) subjects in Chinese universities. This is in line with the government's plans to push for "New Quality Productive Forces", meaning high-tech and more efficient economic development. It has also meant a lower uptake for social sciences.

# Scrutiny and Security: How China Conducts Its Biggest College Entrance Exam

## ● What is the Gaokao?

- China's national college entrance exam (“the highest exam”); over 1.29 crore students appeared this year
- Conducted since 1952; was put on hold during the Cultural Revolution (1966–76)
- Compulsory subjects: English, Mandarin and Mathematics; optional science/social science subjects

## ● Security & Conduct

- Question papers drafted under strict security; isolated locations including military facilities and prisons used for preparation
- Biometric verification at entry to detect student impersonation
- Intelligent security gates detect small metal objects (phones, smartwatches, hidden earpieces); some regions use “AI-powered”, real-time surveillance flagging irregular behaviour
- Grading computers run on an exclusive network to prevent hacking; exam rooms monitored 24/7 by police

## ● Punishments

- A 2016 case saw six teachers handed jail sentences (1 to 26 months) for “illegally obtaining state secrets”

# Biochar Offers a Way to Turn India's Farm Smoke into 'Black Gold'

## Biochar offers a way to turn India's farm smoke into 'black gold'

When added to poor, degraded soils, biochar can increase their ability to store water and stimulate the growth of beneficial microbes; studies have indicated that it can improve crop productivity by 10% to 30% and water-holding capacity by 10% to 25%, particularly in soils low in nutrients

Vinaya Kumar H.M.,  
Vikram Patil

India's agriculture faces a serious paradox: a large amount of biomass that could improve soil health is instead being burned. This threatens soil health and food security. Punjab and Haryana burn more than 20 million tonnes of paddy straw in their open fields every year. This is due to the short post-harvest periods and a lack of practical alternatives.

Burning these residues releases large quantities of greenhouse gases and fine particles into the air and soil, causing serious air pollution in the region as well as a major loss of organic matter that could otherwise be returned to the increasingly depleted soils.

At the same time, large tracts of agricultural land, from the black soils of Maharashtra to the red soils of Kerala, suffer from very low soil organic carbon, poor water-holding capacity, and rapid loss of nutrients, reducing crop productivity even with better seeds and irrigation.

Both these problems are symptoms of the same larger failure to recycle natural resources efficiently.

### Value to agriculture

This is where biochar emerges as a carbon negative solution. Biochar is made by heating agricultural waste in low-oxygen conditions. What remains is a carbon-rich material that breaks down very slowly in the soil, helping to keep carbon locked away for long periods of time. The benefits extend well beyond carbon capture. Biochar is highly porous and contributes to aggregating soil particles, holding water, and creating a suitable environment for microorganisms.

Given these properties, the value of biochar to agriculture should be obvious. When added to poor, degraded soils, it can increase their ability to store water and stimulate the growth of beneficial microbes.

Studies have indicated that it can improve crop productivity by 10% to 30% and water-holding capacity by 10% to 25%, particularly in soils low in nutrients.

Biochar derived from maize stalks and applied to black soils in Akola district of Maharashtra has improved soil organic carbon content and overall soil fertility in field trials. Research from Kerala has shown that biochar made from coconut leaf stalks increased soil quality in different cropping systems, emphasising the need to use local resources effectively. Perhaps most significantly, long-term studies have shown that biochar has the benefit of enhancing soil health and maintaining higher crop output over time.

### Wider vision

This approach also supports India's wider vision of sustainable agriculture and climate resilience. As droughts,



A biochar mixture ready to be applied to soil. Biochar is made by heating agricultural waste in low-oxygen conditions. What remains is a carbon-rich material that breaks down very slowly in the soil. [NABARD/CC-01](#)

heatwaves, and erratic rainfall become more frequent and more intense, improving soil health will be crucial to sustaining agricultural productivity.

By increasing the water-holding capacity and the ability to use nutrients efficiently, thus, biochar can help crops withstand moisture stress while reducing dependency on external inputs. This is particularly important for small and marginal farmers who are often the most exposed to the climate's vagaries.

Integrating biochar into the existing initiatives on natural farming, soil health management, and carbon farming can deliver environmental and economic benefits at scale. Yet in India, biochar remains largely confined to research trials and pilot projects, and is very much alien to many farmers.

In fact, agricultural residues are generally seen only as a problem of disposal. But they are an important resource that can generate additional income, create jobs, can deliver environmental benefits, and could even support payments for ecosystem services.

### Carbon credits

As one way forward, the government could package the activities of turning agricultural waste into biochar and putting it on soil into carbon credit markets, creating a strong economic incentive for mass adoption.

Biochar carbon already passes rigorous scrutiny criteria for long-term sequestration, and it can be classified as a persistent carbon dioxide removal technology under internationally accepted accounting standards. For

**Biochar is highly porous and contributes to aggregating soil particles, holding water, and creating a suitable environment for microorganisms**

instance, VM0042 agricultural land management methodology quantifies both avoided emissions from residue burning and long-term carbon sequestration in soils. In this protocol, each tonne of certified biochar can generate 22.8 t carbon dioxide-equivalent in carbon credits. Depending on carbon market prices, certified biochar can thus provide an additional source of income for project developers, farmers, and cooperatives.

This approach is already being tested in projects such as those using the KISAN kiti from IFP/Kharagpur, which allows smallholders to monetise farm waste.

Several international examples also illustrate the scalability of biochar systems. In Kenya, turning rice husks into biochar has produced thousands of certified carbon credits and improved soil pH and phosphorus content. Thailand has pushed biochar use through national initiatives on soil rehabilitation and carbon management, and tied certification to access to the national carbon registry system, providing a policy-to-market pipeline. High carbon retention and large yield gains have also been reported from Brazil's Embrapa Institute after using on-farm biochar generated from sugarcane bagasse.

These examples also show that success

depends on integrated strategies that combine decentralised, appropriately scaled pyrolysis technology with strong measurement, reporting, and verification systems.

### Urban organic waste

The feedstock for biochar is also not limited to agricultural waste and can encompass urban organic waste as well. India generates around 62 million tonnes of municipal solid garbage per year and more than 50% of that is biodegradable. Sewage sludge and crop residues can also be converted into biochar.

This is in keeping with the precepts of the circular economy, as organic waste is diverted from landfills that produce methane and is converted to a useful product for agriculture. By systematically implementing these measures, India can turn its large waste streams into 'black gold', thus ensuring a more resilient agricultural future while making a meaningful contribution to global climate mitigation efforts.

Overall, the value of biomass can only be realised through an integrated ecosystem that catalyses innovation, entrepreneurship, market linkages, investment, and cost-effective access to biochar for farmers.

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### THE GIST

Biochar is made by heating agricultural waste in low-oxygen conditions

Integrating biochar into the existing initiatives on natural farming, soil health management, and carbon farming can deliver environmental and economic benefits at scale

As droughts, heatwaves, and erratic rainfall become more frequent and more intense, improving soil health will be crucial to sustaining agricultural productivity

In India, however, biochar remains largely confined to research trials and pilot projects, and is very much alien to many farmers

# Biochar Offers a Way to Turn India's Farm Smoke into 'Black Gold'

## ● The Problem

- Punjab and Haryana burn more than 20 million tonnes of paddy straw annually, causing air pollution and loss of organic matter
- Many Indian soils — from Maharashtra's black soils to Kerala's red soils — suffer from low organic carbon, poor water-holding capacity and rapid nutrient loss

## ● What is Biochar?

- Made by heating agricultural waste in low-oxygen conditions (pyrolysis); leaves a carbon-rich material that breaks down very slowly in soil
- A carbon-negative solution locks away carbon for long periods

## ● Benefits to Agriculture

- Highly porous; aggregates soil particles, holds water, supports beneficial microorganisms
- Can improve crop productivity by 10–30% and water-holding capacity by 10–25%, particularly in nutrient-poor soils
- Field trials: biochar from maize stalks improved soil organic carbon in Akola, Maharashtra; biochar from coconut leaf stalks improved soil quality in Kerala

# Biochar Offers a Way to Turn India's Farm Smoke into 'Black Gold'

## QUESTION

GS

Hard

Environment & Ecology

Prelims 2020

What is the use of biochar in farming?

1. Biochar can be used as a part of the growing medium in vertical farming.
2. When biochar is a part of the growing medium, it promotes the growth of nitrogen-fixing microorganisms.
3. When biochar is a part of the growing medium, it enables the growing medium to retain water for longer time.

Which of the statements given above is/are correct?

Select an option to attempt

☐ A 1 and 2 only

☐ B 2 only

☐ C 1 and 3 only

☐ D 1, 2 and 3

# **Thank You**